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Application of interactive sensory arts exhibition in promoting the protection of endangered species: The Elephant tales

Smith Boonchutima *, Sitamon Ratanavadi, Russarin Chaowjirakit, Karnpitcha Prayotamornkul

Chulalongkorn University, Thailand

A R T I C L E I N F O

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A B S T R A C T

The issue of conserving the Thai elephant appears to be a somewhat underrated area, perhaps due to past difficulties in promoting the issue. We hosted an interactive exhibition in a local shopping centre to familiarise visitors with the issue and tested the relationship between perceived interactivity and raised awareness of elephant conservation issues. The exhibition was designed to engage the five human senses. In addition to a wealth of visual elements, we also used sculptures that visitors could touch, forestry scents, elephant noises and ice cream that tasted like fruits elephants prefer. Promotion of the exhibition was done online via social media platforms. The quantitative findings revealed a moderate-to-strong relationship between perceived interactivity and raised awareness, implying that individuals who considered the exhibition more interactive also reported an increase in awareness of the elephant issue. The qualitative findings have shown that one and a half year later participants remembered various interactive parts of their experience and were able to identify different modalities of the presented information. Therefore, the study provided arguments in favour of using multi-modal exhibitions in promoting the protection of wildlife.

1. Introduction

The elephant is one of the largest continental living herbivores present on the red list of endangered species due to drastic habitat loss and environmental issues that affect local ecosystems (Malikhao & Servaes, 2017). According to the World Wildlife Fund (WWF), with human activity as the primary cause, the elephant population has declined by 50% over the last three generations (WWF Indonesia, 2017). It is estimated that 30,000–50,000 Asian elephants live in the wild (Choudhury et al., 2008; Sukumar, 2006), with approximately 15,000 living in captivity (Maurer, Rashford, Chanthavong, Mulot & Gimenez, 2017). In Thailand, it is estimated that 1600−3700 elephants live in the wild, with an additional 3000−4500 elephants held captive, implying that more elephants are held captive than live in the wild (Pintavongs et al., 2014; Sukumar, 2006). The majority of these elephants are used in tourism (Sukumar, 2006), with many of them living in inadequate conditions, lacking adequate social interaction, bathing, food, or the freedom to make any individual choice (Schmidt-Burbach, Ronfot & Srisangiam, 2015; Malikhao & Servaes, 2017). This is in line with findings that elephants held in captivity have higher mortality rates (Clubb et al., 2008). Due to the COVID-19 pandemic, many elephants held in

* Corresponding author at: Faculty of Communication Arts, Chulalongkorn University, Phayatai Road, Patumwan District, Wangmai Subdistrict, Bangkok, 10330 Thailand.

E-mail address: smith.b@chula.ac.th (S. Boonchutima).

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captivity were fed inadequately due to the absence of tourists willing to pay to see them (Hatton, 2020; Paddock & Suhartono, 2020). The cost of feeding an elephant for a day is close to three times the daily minimum wage in Thailand (Hatton, 2020), which makes the entire situation even more complicated for older or unhealthy elephants requiring extra care yet yielding less income to their owners, who are additionally burdened by the economic consequences of the COVID-19 pandemic and the countermeasures applied. However, both wild and captive Asian elephants are an endangered species, which implies that the animals held in captivity require the same effort to address their wellbeing, promote breeding and save them from extinction (Baker & Winkler, 2020; Malikhao & Servaes, 2017).

There are multiple ways in which the public can contribute to the welfare of Thai elephants, for example, by not buying ivory and encouraging others to join conservation efforts (Mathiesen, 2016; WildAid, 2017). However, to engage the public, one has to convey the information efficiently. Although knowledge transfer is an important goal of such communication (Cottrell & Graefe, 1997), it is at least as relevant to arouse positive and action-promoting attitudes on the subject (Ajzen, Joyce, Sheikh & Cote, 2011). To achieve the responsible transmission of information, communication must utilise the correct methods to create a clear, consistent and concise message. Interactive strategies that combine, for example, key visuals, public relations and online promotional materials should be used to enhance effective communication Miller's, 1956. information processing paradigm, in combination with Baddeley's (2010, 2012) models of working memory could be used to explain why combinations of different approaches, that is, approaches focused on different modalities, are more likely to succeed in conveying information.

According to Miller, 1956, see Fig. 1), the concept of sensory input describes how humans perceive their environment. The five senses (smell, touch, hearing, taste and sight) are used to gather this information, which is then stored in the sensory register Baddeley (2010). explored the components of short-term memory in detail and highlighted the role of working memory in the retention of auditory and visual stimuli. According to his model of working memory, verbal and visual stimuli are processed using different cognitive systems: the phonological loop and the visuospatial sketchpad. The phonological loop has sub-systems for speech, signs and lip-reading, whereas the visuospatial sketchpad has sub-systems for visual (e.g. colour, shape), spatial and haptic (e.g. kinaesthetic and tactile) information. All these types of information can be merged in the episodic buffer and transferred to long-term memory. The episodic buffer is also held responsible for merging olfactory and tactile information with audio-visual inputs from the phonological loop and the visuospatial sketchpad, although cognitive models for processing these modalities are less well-developed and still speculative (Baddeley, 2012). However, multiple arguments in favour of the relationships between odours and memories (Herz, Eliassen, Beland & Souza, 2004; Tromelin, 2016), taste and memories (Kobayashi et al., 2004; Morris & Dolan, 2001; Okamoto et al., 2011) and touch and memories (Delong, Wu & Bao, 2007; Kinnunen & Kolehmainen, 2019) exist, especially for their role in forming social relationships. If the information is not attended to, that is, not processed by the relevant sub-system of long-term memory, the probability of accessing that information after a prolonged period is negligible (Baddeley, 2010, 2012).

The process of conveying information based on the combination of the above-mentioned principles is known as multi-modal learning (Moreno & Mayer, 2007). Originally, the process was defined as learning enhanced by the use of two modes of conveying knowledge: verbal and non-verbal (Paivio, 2006), which gained scientific support (Fletcher & Tobias, 2005; Moreno, 2006). However, in addition to combining multiple modes, content can be presented in multiple modalities (e.g. auditory, visual, olfactory), which enables it to utilise multiple cognitive systems and enhance the process of learning. The logic behind these principles served as a foundation for the development of the cognitive-affective theory of learning with media (CATLM; Moreno, 2005). The theory postulates that different stimuli are processed by different cognitive systems, and each system on its own has a limited capacity. Therefore, to allow a large amount of information to be processed, it is more effective to present this information in multiple modes and modalities (Moreno & Mayer, 2007). Otherwise, limited attention or interference may prevent adequate processing and undermine the learning outcomes. This is also reflected in the modality principle of instructional design (Low & Sweller, 2005; Moreno, 2006), which states that the most effective learning environments are environments wherein information is presented in multiple modes and modalities. Altogether, cognitive and learning theorists appear to concur on the beneficial effects of utilising multi-modal stimuli in conveying information. All this can be further enhanced by adding interactivity into the learning process, which turns learners into active creators of knowledge rather than passive receivers of information (Moreno & Mayer, 2007), and this insight is also in line with situational learning theories.

However, some drawbacks of multi-modal learning and interactivity have also been noticed. As Moreno and Mayer (2007) highlighted, too much interactivity could undermine the process of learning, especially if the interactions are not focused on creating knowledge. This, in turn, can also mean that some interactions may draw attention away from materials and lead to undermining the

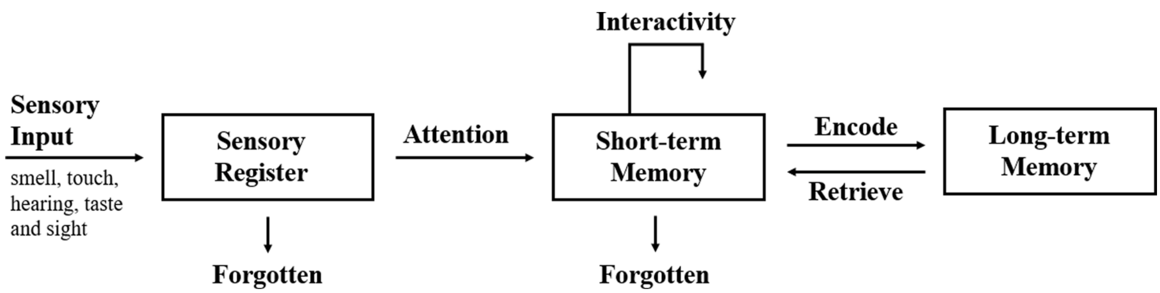

Fig. 1. *Adapted from* Miller's (1956) information processing paradigm.

awareness of concepts. Similarly, distracting stimuli can draw attention away from the concepts and undermine the process of learning (Cohen, Ivry & Keele, 1990; Hanley et al., 2017). However, having the opportunity to interact willingly with objects at an individually adapted pace may aid learners to adapt with the level and type of interactions according to their needs, thereby mitigating the mentioned distractive effects.

The use of multi-modal learning has already been demonstrated in biology teaching (Bull & Dupuis, 2014), chemistry teaching (McDermott & Hand, 2013) and in promoting the protection of cultural heritage (Potočnik, 2017). Therefore, it could be expected that the effectiveness of these methods can be extended to learning about the conservation of endangered species—in the context of this study, the Asian elephant. This led to the development of an interactive exhibition named 'The Elephant Tales', which used all five human senses to raise awareness of Thai elephant conservation. In line with the presented theories, our hypothesis was that perceived interactivity enhances learning, which leads to better awareness of the issue. Therefore, to estimate the short-term relationship between interactivity and learning, we focused on testing the relationship between the perceived level of interactivity and raised awareness, basing our study on Miller's information processing paradigm. Finally, we conducted interviews with six visitors of the exhibition a year and a half after the exhibition to check whether they remember the exhibition and which aspects of the exhibition they remember in order to observe the potential of interactive exhibition to produce long-term changes (at least on the level of opinions). We hypothesized that participants would remember the exhibition and be able to name multiple objects present there and describe their experience using multiple modalities, as well as that most of them would be able to name the purpose of the exhibition and multiple interactive objects that were presented.

2. Study 1

2.1. Methods

2.1.1. Ethics statement

Ethical aspects of both parts of this study were evaluated by the communication management program committee at the Faculty of Communication Arts, Chulalongkorn University, Thailand, and were declared adequate.

2.1.2. Participants

The post-test-only research design was implemented. Of approximately 300 visitors, 231 individuals completed the questionnaire. The sample consisted of slightly more women ($n = 119$) than men ($n = 112$). It was also revealed that most participants ($n = 126$), or 54.5%, were students between the ages of 19 and 27, which might have been due to the mall being located about 500 m from a university.

2.1.3. Measures and analyses

After visiting the exhibition, attendees were asked to rate on the following two parameters in questionnaires: (1) How much interaction they had during the exhibition, and (2) How much awareness they gained regarding elephant conservation and accompanying problems. A Likert-type scale ranging from 1 to 5 was used to rate these questions, with higher values indicating higher numbers of interactions and gained awareness. Data were analysed after the end of the exhibition in R (R Core Team, 2020).

2.1.4. Intervention design and implementation

2.1.4.1. Target audience. The proposed target audience was university students living in Bangkok, between the ages of 18 and 25. This is the cohort that tends to spend a large amount of time on social media platforms where they express opinions and emotions, especially using Instagram (Anderson & Jiang, 2018) and Facebook (Kleinschmit, 2019). Therefore, to promote the exhibition successfully amongst the desired audience, Instagram was utilised. A total of 30 influencers (with more than 5000 followers each) were asked to promote the event on their pages, as recommendations from influencers are trusted more (Jin, Muqaddam & Ryu, 2019) and have more salience than traditional advertising methods (Djafarova & Bowes, 2021).

2.1.4.2. Location and timing. Permission was granted for the use of the 50-square-metre event area in a local community mall connected to the underground train within walking distance from the university, where most of the customers are university students and workers. Admission to the exhibition was free of charge. Speakers were set up in each corner and the space was divided into four sections. At the entrance was a registration table with a photo booth. The three art sculptures (one for each life stage of the elephant—baby, adolescent and adult) were placed diagonally across the area. At the end, information boards were placed, and an area where ice cream was handed out was located around the right corner. The three elephant sculptures were used as the key communication visuals, which contained all of the necessary information (i.e., key problems elephants faced, brief mention of organizations that help elephants, and how citizens could help them in protecting elephants). The installation of the exhibition was done the night before the first day (12 March 2020) and seven volunteers helped with this process. The exhibition elements were placed according to the original floor plan with minor adaptations to optimise the benefits from the facility. Parking coupons were handed over to the exhibition staff for distribution to visitors. As Thai National Elephant Day is on 13 March, the exhibition was held from 12 to 13 March.

2.1.4.3. The five senses exhibition. In line with the multi-modal interactive approach (Moreno & Mayer, 2007) and the presumption of the limited capacity of each sensory channel on its own (Sweller, 1999), a multi-modal interactive exhibition was prepared to enhance comprehension of the situation of Asian elephants in Thailand Miller's (1956). information processing paradigm was used in planning the exhibition to maximise information retention Figs. 2–5. present photos taken during the exhibition.

When developing content for the exhibition, we focused on five sensory modalities: vision, hearing, taste, smell and touch. First, the stimuli focused on attracting visual attention were prepared in the form of three contemporary art pieces done by a well-known local graffiti artist. These art pieces were of different sizes. The smallest represented a young elephant, a medium one represented an adolescent elephant and the largest represented an adult elephant. Six information boards with accompanying leaflets were also made available at the exhibition. The content of these started with general facts, then moved on to the current situation, solutions to these problems and lastly, notes from elephant experts. Participants were allowed to touch the elephant sculptures, while the tactile sense was also used during a live painting session on the second day (13 March 2020) of the exhibition. Thai elephant sounds were recorded at one of the Thai Elephant Conservation Centres. These sounds, which included snorts, grunts, trumpets and chewing, were played as ambient noises during the exhibition. To engage the sense of smell, the location was supplied with a diffuser that released a forestry scent, imitating the natural habitat of elephants. The sense of taste was incorporated by serving fruit-flavoured ice creams with banana, watermelon and pineapple flavours, in line with the diet of Thai elephants. These sensory stimuli were incorporated in the Goldblatt, 2005 Event Management Process, a five-step (research, design, planning, coordination and evaluation) processual model used to plan and execute the exhibition.

2.1.5. *Procedure for conducting the exhibition*

Before the exhibition started, it was promoted on social media platforms (Instagram and Facebook). These platforms were also used to create, track and evaluate the activities of the campaign. Content was posted on both social media sites, which had the purpose of informing the audience on relevant aspects of the exhibition and encouraging discussion and sharing.

Once all sanitising procedures had been adhered to, the attendees were requested to register and were provided with a leaflet. Attendees next moved to the photo booth area and selected props to use in taking their photos. After photographing, attendees were encouraged to tour the elephant sculptures, starting at the youngest (smallest) before moving on to the adolescent and then to the oldest (largest). These elephants were placed in the middle of the exhibition and represented three groups of elephants that were inadequately protected: orphan elephants, injured elephants and old elephants. The six information boards were then encountered. Once they had finished reading the boards, attendees were encouraged to scan the QR code found on the leaflet so they could complete the survey focused on evaluating the exhibition. Finally, a mini-game was played to determine which flavour of ice cream (banana, watermelon and pineapple) each attendee would receive. In the end, attendees returned to the photo booth to receive their printed photos, as well as a digital version that was sent to their mobile devices.

2.1.6. *COVID-19 prevention*

Due to the COVID-19 pandemic, the exhibition area had to be sanitised before any elements of the installation could be unpacked, prepared and placed. These included boards, standees, loudspeakers, freezers and tables. A protection plan was developed according to the Centers for Disease Control and Prevention's guidelines (CDC, 2020a, b). The COVID-19 protection plan consisted of five steps.

Fig. 2. Visitors took photos holding the funny and informative messages. Then, they received photo stickers as souvenirs and the event organizers made copies to affix at the exhibition's boards.

Fig. 3. Visitors listening to the explanation from the event organizers and looking at the textual information on the board.

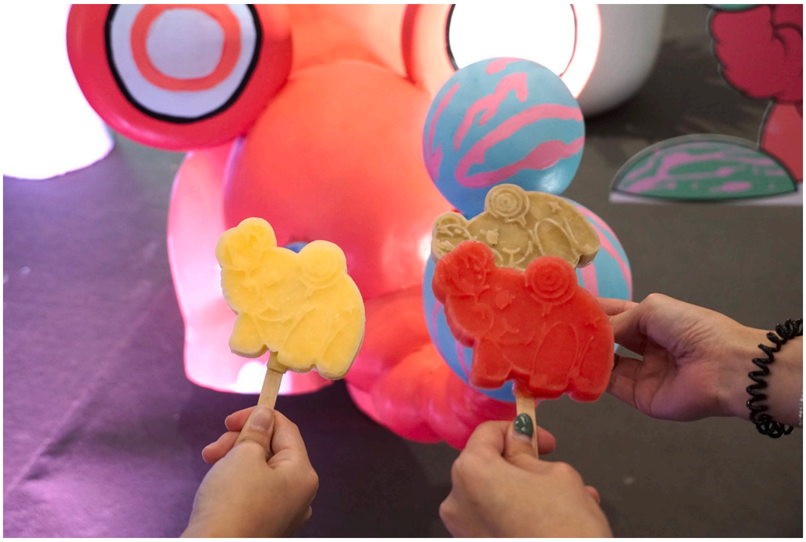

Fig. 4. Visitors got elephant-shaped popsicles made from pineapples, watermelons, or bananas.

First, bottles of alcohol-based gel were made available for attendees at the registration table. Staff members with bottles of alcohol-based gel were also placed in each corner of the exhibition area. Second, a staff member was responsible for checking each patron's temperature at the registration table. The temperature had to be below 37 °C for the attendee to be granted access. Third, each visitor had to sanitise their hands at the registration table. Next, the disinfectant spray was utilised to sanitise all public surfaces that were likely to be touched. Lastly, all staff members were required to wear face masks for the entire duration of the exhibition.

2.2. Results

This section briefly presents the outcomes of the evaluation of the exhibition. First, the descriptive data on interactivity and exposure are presented, followed by their correlation.

Table 1 suggests that the majority of attendees considered the exhibition to be appropriately interactive and noticed a rise in their awareness of elephant conservation problems. Moreover, the fact that none of the attendees selected the 'very low' option as a

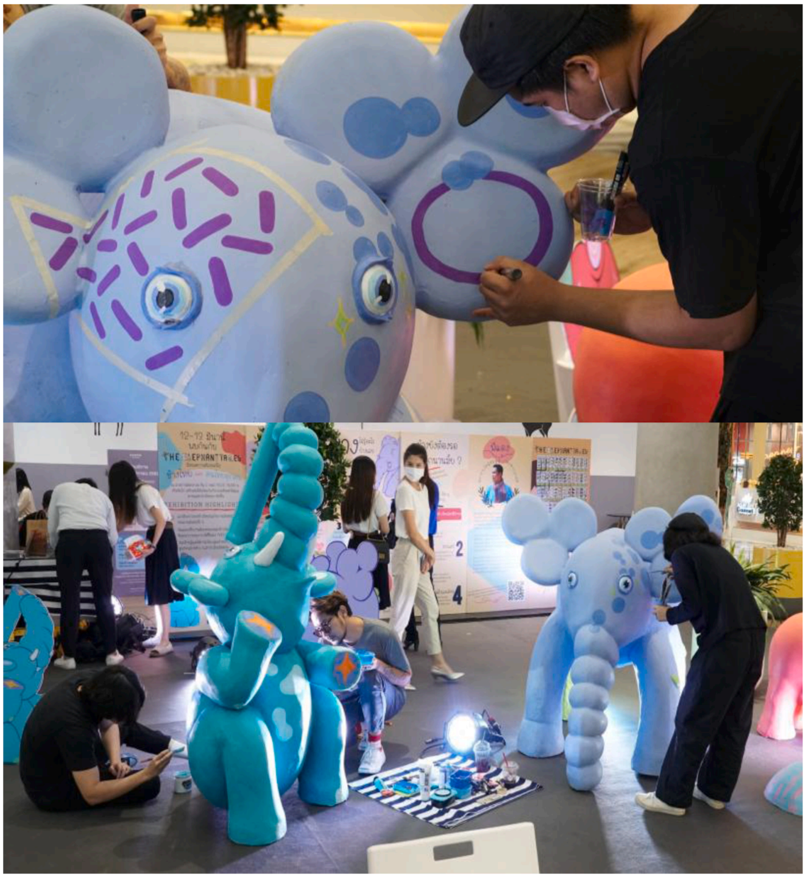

Fig. 5. The artists painting the elephant sculptures to depict the conditions of an ageing elephant: wounds, scars, and blindness.

Table 1
Frequencies and descriptive statistics of perceived interactivity of the exhibition and raised awareness of Thai elephants amongst attendees ($N = 231$).

Variable	1 (very low)	2 (low)	3 (moderate)	4 (high)	5 (very high)	*M*	*SD*
Perceived interactivity	0	0	12	54	165	4.68	0.55
Raised awareness	0	2	4	60	165	4.66	0.57

description of the change in their awareness implies that the applied intervention was generally effective (considering the limitations of the underlying research design).

The Pearson correlation coefficient was calculated to estimate the relationship between perceived interactivity and effectiveness in raising awareness. The obtained value of r was equal to 0.64 ($p < 0.001$, 95% CI[0.56, 0.71]), indicating that participants who considered the exhibition as being more interactive on average also reported a more substantial rise in their awareness of elephant conservation problems. However, the distribution of both perceived interactivity and raised awareness seem to be skewed, which violates the normality assumption, one of the relevant preconditions of Pearson's coefficient of correlation (Field, 2013). Additionally, none of the two variables was continuous; their distribution was categorical. Therefore, we also calculated the Kendall tau correlation coefficient, robust to these deviations. The results were nearly identical ($\tau = 0.62$). To check the robustness of these findings, we conducted another study one and a half years later in order to explore the long-term effects of multi-modal exhibitions.

3. Study 2

3.1. Methods

3.1.1. Participants

Interviews were conducted with six of the visitors of the exhibition "Elephant tales". One visitor was a woman, while the other five were men. On the average, interviewees were between 24 and 25 years old.

3.1.2. Measures

Six broad questions were used in the semi-structured interviews. The questions were formed based on the principles of cognitive interviews in order to maximize the recall prior to providing any information that might serve as help in remembering the exhibition. By doing so, we observed the contents that our interviewees were likely to convey to anyone who asked them about the exhibition.

1 Please describe your walk through the exhibition, from start to the end.
2 Please describe your experiences backwards - from the end to the start. This may help you to remind yourself of some additional details.
3 Please go from the beginning to the end once again. This might help incorporate details to what you remembered when describing the process backwards.
4 Please specify the object from the exhibition you remember the most and why you think you remember it.
5 Please describe specific objects that you were exposed to in the exhibition. Ice cream, sculptures, boards, sticker station, sounds, the crowd, and elephant poo. What were they like?
6 From visiting the exhibition, please describe what problems of the Thai elephants you remember the most and why you think you remember those.

3.1.3. Procedure

The semi-structured cognitive interviews were conducted over telephone calls, during the last week of November 2021. When visiting the exhibition, visitors had the option to leave personal contact information in case they were willing to participate in follow-up studies or interested in finding out more about the subject. Fifteen participants left their contact numbers. They were contacted in the middle of November 2021 and asked if they were willing to participate in a follow-up interview. Six visitors who were able to participate in interviews in one of two available dates were interviewed. Prior to participating in the interviews, participants were reminded of their rights, encouraged to provide honest answers and informed on their anonymity and that information provided in the interview, even if negative, will not cause any material or non-material damage to the team that prepared the exhibition. Interviews were conducted in Thai and their transcription has been translated to English.

3.2. Results

The outcomes of cognitive interviews provided arguments in favour of the value of interactivity. When describing their experience, all interviewees mentioned visual (images and sculptures of elephants, props and photos), and gustatory (ice cream) stimuli experienced during the exhibition. Although five out of the six interviewees mentioned auditory stimuli, two mentioned hearing the voices that narrated the story, while three mentioned elephant sounds. Only one participant mentioned olfactory stimuli (smell of eucalyptus). However, all the participants mentioned interacting with some elements of the exhibition. Altogether, although none of the interviewees mentioned all the details regarding the exhibition and each interviewee described different elements of the exhibition, all of them described a multi-sensory experience and remembered fairly well the subject of the exhibition.

Interviewees were also asked to name a few objects from the exhibition they remembered most clearly and try to describe them. Generally, participants had no problem with completing such a task, with most of them describing the elephant sculptures or paintings.

Then, we asked participants why they remembered the elements of the exhibition they remembered. One interviewee did not provide any answer, while two pointed out the distinctiveness of the central elephant sculpture as the main reason for remembering it.

> *Elephant statue, uh, because I remember because, as I said earlier, it's set in a fair, and it's quite prominent and it's a large statue of a certain level. (Male visitor, age 24)*

> *Elephant statue, because it's the most prominent, and it's a strange elephant, not quite like anywhere else. (Male visitor, age 25)*

> *Because, at that time, I was studying art, and the statue was quite large, and I understood that the way to make the statue of that scale was quite time consuming and costly, so I went back and looked at it especially, and I wanted to pay attention to the artists. I wanted to see how they played with colour … (Female visitor, age 24)*

Another interviewee focused on the uniqueness of the combination of simultaneously presented stimuli, which enhanced his experience and, consequently, allowed him to better remember the contents.

> *I think it's something that's unique, like the statue, I remember it because I saw it and it was eye-catching, and the sound and the smell, and I felt that other exhibitions weren't like this, like if we walked through museums like this, it would have smelled like aroma in a room, right? But this one was kind of like, like, it's like they happened at the same time, both smells and sounds, so I remember well. (Male visitor, age 27)*

An interviewee made a similar comment related to a painting of an elephant.

The remaining interviewees mainly spoke about remembering objects they had the opportunity to interact with.

> *Like, uh… We could engage with those objects, you know, it's action. There's a kind of surface to touch that helps us remember things better. Kind of more than just reading and walking past. Something like that. (Female visitor, aged 24)*

> *Because I lifted the statue and I saw that people like to take pictures with the statues… (Male visitor, aged 24)*

Finally, participants generally remembered the main points of the exhibition – deforestation implying loss of natural habitats for wild elephants and lack of care for elephants kept in captivity. Specific examples of comments they made are as follows.

Kind of living space conflict? Yeah, this one I remember. (Male visitor, aged 24)

Well, I remember that, uh, the problem of elephant, most of the problems come from people. It is like people invading their area, and then, uh, we're actually the ones who go into their areas. We're the ones who go and destroy their area first. (Female visitor, aged 24)

I think the problem with Thai elephants is the lack of care. I can't remember the words exactly. But the point that I remembered was that I understood that it was lacking of attention, and yeah, it lacked attention, and I thought that when we looked into it, we felt that Thai elephants had been with us for a long time, that they had been with Thai society for a long time, and that the elephants are major symbols in Thailand. We, as a new generation, have overlooked the importance of elephants. (Male visitor, aged 27)

4. Discussion

This combination of an interactive exhibition and a study allowed us to test the premises of multi-modal learning. In general, visitors of the exhibition reported increased awareness of the elephant conservation issue and a high level of interactivity during the exhibition. Even a year and a half later, interviewed visitors remembered many details of the exhibition and its main purpose. This is in line with results of earlier studies (Bull & Dupuis, 2014; McDermott & Hand, 2013) that confirmed the premises of CATLM (Moreno, 2005; Moreno & Mayer, 2007) that when people are provided with multiple emotional and cognitive stimuli, especially if they have the option to interact with no external pressure similar to our study, the likelihood of raising awareness on a specific issue, which also represents a form of learning, is increased. Presenting content in multiple modes and modalities may activate different cognitive mechanisms and result in a more thorough information processing (Baddeley, 2010), thereby increasing the likelihood of remembering the content. Although causal relationships cannot be established using this type of research design, it provides another argument in favour of the conclusions of earlier studies that highlighted the positive role of interactivity and multiple modalities in learning.

Participants who perceived the exhibition as more interactive also expressed higher awareness of the elephant conservation problems. That all participants expressed at least somewhat higher awareness of these problems highlights the potential effectiveness of interactive exhibitions in evoking changes in public opinion, and as in this case, promoting animal conservation Brych (2018, p. 22). concluded that 'application of the multimedia exhibition technologies is becoming a promising direction of the museums' development today' Chiu. created a multimedia book where readers were introduced to the issue of environmental protection Kuhar, Bettinger, Lehnhardt, Tracy and Cox (2010). discovered long-term retention of information related to conservation after implementation of a conservation-related educational programme, while a combination of films and educational magazines that was used with children in Uganda motivated them to encourage the conservation of great apes (Slavin, 2014). The findings from our cognitive interviews support these earlier findings: visitors generally remembered the most relevant aspects of the exhibition, with most of them being able to vividly recall the multiple modalities of presented stimuli. While the majority of interviewees did not provide any scientific explanation of why they remember the specific aspects of the exhibition, one interviewee mentioned that interactivity of the exhibition helped her to retain the content. The fact that multiple modalities of content were remembered goes in line with the narrative basis of episodic memory as a component of declarative long-term memory (Baddeley, 2001): participants were instructed to mentally revisit the exhibition, which allowed the stimuli of various modalities to serve as cues for remembering other stimuli. Their narratives may be the strongest arguments in favour of the relationship between interactivity and learning: while describing the exhibition, all the interviewees described their own options and activities or activities of others, all of which may represent responses to specific stimuli presented at the exhibition. It was interesting to note that even the interviewee who did not remembered many details about the exhibition remembered his activities during the exhibition and objects he considered peculiar. In line with the notion of episodic memory, our study provides arguments in favour of careful organization of exhibitions: it seems important to organize exhibitions in the form of easy-to-remember stories rather than just presenting contents without highlighting their functions and relationships. Multiple interviewees remembered the elephant sculptures, designed to attract attention. This is in line with the von Restorff effect (von Restorff, 1933), according to which people tend to remember stimuli they consider different from the majority of others. This could also have practical implications: most relevant ideas could be presented at a dominant place in an exhibition and in a visually expressive manner in order to be remembered in the long run.

However, as Manubay et al. (2002) concluded, the effects of such exhibitions should not be over-generalised. While exhibitions focusing on environmental protection may motivate individuals to start thinking about these problems or discussing them with peers, they may not be as efficient in changing behaviours. Together with the outcomes of this study, three conclusions can be drawn: first, interactive and multimedia exhibitions are an effective way of increasing public awareness of some issues; second, messages conveyed using such exhibitions can be remembered for a long period after the exhibition; and third, further studies are required to determine the extent of the possible resultant behavioural effects of these remembered messages. Such studies should take both individual and contextual factors into account in order to serve as stepping stones towards developing more efficient interventions.

Several limitations of this study should also be considered. First, convenience sampling implies that caution should be exercised when generalising these results. Of all the people who witnessed the notice about The Elephant Tales, only a sub-section actually visited the exhibition, and of all the visitors, only a sub-sample completed the survey. As a result, the sample that completed this study was made up of individuals who came to the exposition, were willing to participate in a short survey and were already interested in elephant problems. Furthermore, only a small subsample of that subsample took part in the interviews, which might have biased the results of the qualitative part of our study. Although pre-test and post-test experimental designs usually serve as a gold standard for determining if an intervention was successful or not, this was impossible in our case due to the nature of the exhibition. However, the way we framed our questions allowed us to capture a subjective difference in awareness, with almost all the participants expressing at least some, and the majority expressing substantial perceived awareness gained. Nevertheless, we recommend future researchers to use multi-item measures of relevant concepts and experimental design, possibly on more representative samples and behavioural outcome measures, to confirm the robustness of these findings.

Despite the study's shortcomings, it can serve as a good starting point for further research on the learning potential of interactive exhibitions. Scientific confirmation of the learning benefits provided by this kind of exhibition could provide a new and effective method of conveying information and raising awareness not only amongst animal conservation activists but also amongst lecturers in general. In future studies, it would be interesting to see how specific combinations of modalities affect learning, as this may be used to adapt educational environments to enhance the process of learning.

5. Conclusions

Altogether, the study provided arguments in favour of the long-term retention of interactive contents. Firstly, it revealed that participants who considered the exhibition to be more interactive also were more likely to report increased awareness of the elephant issue. Secondly, it revealed that participants remembered many objects and activities a year and a half after the exhibition, which is especially significant as they were not instructed to pay special attention to the exhibition. Thirdly, interviewed respondents remembered multiple modalities of experienced stimuli, which may have helped them to relive the experience. Altogether, the use of interactive and multi-modal content in conveying information about endangered species seems to be an efficient way of raising awareness. Therefore, the study provides arguments in favour of the existing approaches that recognise the role of multi-modality in learning.

6. Author statement

Smith Boonchutima: Conceptualization, Methodology, Supervision, Validation, Visualization, Investigation, Writing- Original draft preparation, Reviewing and Editing. **Sitamon Ratanavadi**: Methodology, Exhibition organizing, Data curation, Writing- Original draft preparation, Project administration, Resources. **Russarin Chaowjirakit**: Methodology, Exhibition organizing, Data curation, Formal analysis, Writing- Original draft preparation. **Karnpitcha Prayotamornkul**: Methodology, Exhibition organizing, Data curation, Formal analysis, Writing- Original draft preparation.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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